Lecture flow

Sr. No.	Course Name	Expected Duration (Week)	Expected Duration (Hr)
1.	Core Java	6 Week	55+
2.	Basic MySQL	1 Week	06
3.	Advance MySQL	2 Week	10
4.	JDBC	1 Week	06
5.	HTML	1 Week	06
6.	Basic & Advance CSS	2 Week	15
7.	Bootstrap	1 Week	09
8.	Basic & Advance JS	2 Week	12
9.	JQuery	1 Week	06
10.	React JS	2 Week	15
11.	Servlet & JSP	3 Week	20
Total		**22 Week** (5 Month+)	**160+ hr/Session**

Complete Industrial Core Java Syllabus and Sub Topics

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1. Basics of Java

Chapter 1 contains the basic introduction to the Java language such as

√ What is Java?

√ History and Features of Java

√ C++ vs Java

√ Hello Java Program

√ Internal How to set the path?

√ JDK, JRE, and JVM (Java Virtual Machine)

√ JVM Memory Management

√ Internal details of JVM

√ Unicode System, Operators, Keywords, and Control Statements like if-else, switch, For loop, while loop, etc.

2. Class, Object, and Types of classes

Chapter 2 deals with the most important and core concepts of Java. They are:

√ Naming convention of Java

√ Classes, Objects, and Features. It explains how to declare a class, how to create an object in Java.

√ Object declaration and initialization

√ Life cycle of an object

√ Anonymous object in Java

3. **Packages in Java**

Chapter 3 deals with Packages in Java. Under this chapter, we will learn the following topics.

√ How to declare package in a company project

√ Package naming conventions

√ Sub packages

√ Types of packages such as user-defined packages, built-in packages

√ Importing packages in Java

4. Data types in Java

This chapter deals with the following topics in Java.

√ Data types in Java

√ Primitive data types

√ Non-primitive data types

√ Memory allocation of primitive and non-primitive data types, etc.

5. Variables, Constants, and Literals

Chapter 5 discusses three topic variables, constants, and literals. You will learn the following subtopics in this chapter.

√ Variable declaration & initialization

√ Naming convention

√ Types of variables such as local variables, instance variables, and static variables

√ Scope and memory allocation of variables.

6. Methods in Java

√ Methods in Java

√ Use of method in Java

√ Method declaration, method signature

√ Types of methods in Java: predefined method, user-defined methods: instance method, static method

√ Calling of method

√ Java main method

√ Return type in Java.

7. Constructor in Java

In this chapter, you will familiar with topics like:

√ What is Constructor in Java?
√ Types of constructors: Default and Parameterized constructors

√ Java constructor overloading

√ Constructor chaining in java

√ Copy constructor in Java

8. Modifiers in Java

This chapter deals with topics like

√ What is Access modifier and Non-access modifier in Java?
√ Types of access modifiers like private, default, protected, and public

√ Types of Non-access modifiers like abstract, final, native, static, Strictfp, synchronized modifier, transient, volatile.

9. Static Keyword

This chapter deals with the following important topics.

√ What is Static keyword?

√ Static variable
√ Static method

√ Static block, Instance block

√ Static Nested Class in Java

√ Difference between static variable and instance variable, static method and instance method, static block, and instance block.

10. Final Keyword

Under this chapter, you will learn three important topics:

√ Final keyword
√ Final variable

√ Final method

√ Final class.

11. Super and this Keyword

This chapter can be partitioned into two sections: Super and This keyword. The first section discusses with

√ Super keyword
√ Calling of superclass instance variable

√ Superclass constructor

√ Superclass method.

The second section deals with

√ This keyword

√ Calling of current class constructor, and method.

12. OOPs concepts

In this chapter, you will learn the most important topic Object-oriented programming system (OOPs). In the OOPs concept, you will learn class, object, encapsulation, inheritance, polymorphism, and abstraction. All topics are very important for interview purposes.

13. Encapsulation

This chapter deals with the following topics in Java.

√ Encapsulation in Java

√ How to achieve Encapsulation

√ Data hiding

√ Tightly encapsulated class

√ Getter and setter method in Java

√ Naming convention of getter and setter method

14. Inheritance

This chapter deals with

√ Inheritance in Java
√ Is-A Relationship

√ Aggregation and Composition(HAS-A)

√ Types of inheritance: Single level, Multilevel, Hierarchical, Multiple, and Hybrid inheritance.

15. Polymorphism

This chapter deals with

√ Polymorphism in Java,
√ Types of polymorphism: Compile-time polymorphism and Run-time polymorphism

√ Static and Dynamic Binding

√ Method overloading
√ Method overriding
√ Rules of method overloading and method overriding, various example programs related to rules of overloading and overriding.

√ Covariant Return type

16. Abstraction

In this chapter, you will familiarize with the most important topics in Java.

√ Abstraction in Java
√ Abstract class

√ Abstract method

√ Interface in Java
√ Nested interface, rules, and example programs.

17. Garbage Collection

This chapter deals with garbage collection in Java.

18. Input Output Stream

In this chapter, you will learn topics like

√ FileOutputStream, FileInputStream

√ BufferedOutputStream, BufferedInputStream

√ SequenceInputStream

√ ByteArrayOutputStream, ByteArrayInputStream

√ DataOutputStream, DataInputStream

√ Java FilterOutputStream, Java FilterInputStream

√ Java ObjectStream, Java ObjectStreamField

√ Console

√ FilePermissionWriter, Reader, FileWriter, FileReader

√ BufferedWriter, BufferedReader

√ CharArrayReader, CharArrayWriter

√ PrintStream, PrintWriter

√ OutputStreamWriter, InputStreamReader

√ PushbackInputStream, PushbackReader

√ StringWriter, StringReader

√ PipedWriter, PipedReader

√ FilterWriter, FilterReader, File FileDescriptor, RandomAccessFile, and java.util.Scanner.

19. Collections Framework

This chapter is one of the most favorite chapters of the interviewer. It deals with the following important topics like

√ What is Collections Framework?
√ List, Set, SortedSet, Queue, Deque, Map, Iterator, ListIterator, and Enumeration.

√ ArrayList, LinkedList, HashSet, LinkedHashSet, TreeSet, ArrayDeque, PriorityDeque, EnumSet, AbstractCollection, AbstractList, AbstractQueue, AbstractSet, and AbstractSequentialList.

√ Map, Map Entry, SortedMap, and NavigableMap

√ HashMap, LinkedHashMap, TreeMap, IdentityHashMap, WeakHashMap, and EnumMap.

√ Comparator, RandomAccess interfaces as well as Observable class.

20. Serialization

This chapter deals with Serialization, Deserialization, and Java transient keyword.

21. Exception Handling in Java

This chapter is very important for any Java technical test or Java technical interview. In this chapter, you will familiar with

√ Exception Handling in Java
√ Try-catch block

√ Multiple Catch Block

√ Nested try block

√ Finally block

√ Throw Keyword

√ Throws Keyword

√ Throw vs Throws, Final vs Finally vs Finalize

√ Exception Handling with Method Overriding Java Custom Exceptions

22. Java Annotations

This chapter deals with Java annotations, Built-In Java annotations like @Override, @SuppressWarnings, @Deprecated, @Target, @Retention, @Inherited, @Documented, Java custom annotations, and types of annotations.

23. Java Array

This chapter deals with

√ Java Array

√ Types of array: single dimensional array, multidimensional array, declaration, instantiation, and initialization of Java array

√ Passing array to a method

√ Anonymous array in Java

√ Cloning an array in Java

24. String, String Buffer, String Builder

This is the most important chapter in the whole core java. It will deal with basically three topics such as

√ String,

√ Immutable String

√ String Comparision, String concatenation

√ Substring

√ StringBuffer class

√ StringBuilder class

√ toString method

√ StringTokenizer class

25. Java Thread

√ Java multithreading

√ Multithreading life cycle of a thread creating

√ Thread scheduler

√ Sleeping a thread, Start a thread twice

√ Calling run() method

√ Joining a thread

√ Naming a thread

√ Thread priority,

√ Daemon thread

√ Thread pool

√ Thread group

√ Shutdownhook

√ Java Synchronization: synchronized method, synchronized block, static synchronization

√ Deadlock

√ Inter-thread Communication

√ Interrupting Thread

JDBC Syllabus -

This chapter deals with

√ JDBC Drivers

√ Steps to connect to Database

√ Connectivity with MySQL

√ Connectivity with Access without DSN

√ DriverManager

√ Types of JDBC statements: Statement, Prepared statement, Callable statement

√ Database Metadata, Resultset Metadata

√ ResultSet, types of ResultSet,

√ Storing image, Retrieving image

√ Storing file, Retrieving file, Stored procedures, and functions

√ Transaction Management

√ Batch Processing

√ JDBC New Features, Mini Project, and interview questions.

Basic & Advanced MySQL Syllabus –

MYSQL DATA DEFINITION using SQL

- Manage Databases
- MySQL Table Types
- MySQL Data Types
- MySQL CREATE TABLE
- MySQL Primary Key
- MySQL Foreign Key
- MySQL Sequence
- MySQL INT Data Type
- MySQL DECIMAL Data Type
- MySQL DATE Data Type
- MySQL TIME Data Type
- MySQL DATETIME Data Type
- MySQL ALTER TABLE
- MySQL RENAME TABLE
- MySQL ADD COLUMN
- MySQL DROP TABLE
- MySQL Temporary Table

Basic DATA MANIPULATION using SQL

- SQL Syntax
- What is DDL, DML and DCL?
- SQL Select
- SQL Distinct
- SQL Where
- SQL And & Or
- SQL Order By
- SQL Insert Into
- SQL Update
- SQL Delete
- SQL Injection
- SQL Select Top
- SQL Like
- SQL Wildcards
- SQL In
- SQL Between
- SQL Aliases
- SQL Joins
- SQL Inner Join
- SQL Left Join
- SQL Right Join
- SQL Full Join
- SQL Union
- SQL Select Into
- SQL Insert Into Select
- SQL Create DB
- SQL Create Table
- SQL Constraints
- SQL Not Null
- SQL Unique
- SQL Primary Key
- SQL Foreign Key
- SQL Check
- SQL Default
- SQL Create Index
- SQL Drop
- SQL Alter
- SQL Auto Increment
- SQL Views
- SQL Dates
- SQL Null Values
- SQL Null Functions
- SQL Data Types
- SQL DB Data Types

SQL Functions

- SQL Avg()
- SQL Count()
- SQL First()
- SQL Last()
- SQL Max()
- SQL Min()
- SQL Sum()
- SQL Group By
- SQL Having
- SQL Ucase()
- SQL Lcase()
- SQL Mid()
- SQL Len()
- SQL Round()
- SQL Now()
- SQL Format()

MYSQL STORED PROCEDURES -

- MySQL Stored Procedures Introduction
- MySQL Stored Procedures Getting Started
- MySQL Stored Procedure Variables
- MySQL Stored Procedure Parameters
- MySQL Stored Procedures – Return Multiple Values
- MySQL IF Statement
- MySQL CASE Statement
- IF vs. CASE
- MySQL Loop Statements
- MySQL Cursor
- MySQL Stored Procedures Listing
- **MySQL Error Handling**
- MySQL Stored Procedures – Raising Error Conditions
- MySQL Stored Function

INTRODUCTION TO SQL TRIGGER

- MySQL Triggers Implementation
- Create Trigger in MySQL
- Managing Triggers in MySQL
- Create Multiple Triggers For The Same Trigger Event And Action Time
- Working with MySQL Scheduled Event
- Modifying MySQL Events

MYSQL VIEWS

- Introduction to Database View
- Views in MySQL
- MySQL CREATE VIEW
- MySQL Views & WITH CHECK OPTION
- Creating MySQL Updatable Views
- LOCAL & CASCADED in WITH CHECK OPTION
- Managing Views in MySQL

HTML Syllabus

Module 1: Web Programming Introduction

- Architecture of a website
- Different technologies in making the website
- Web Development Introduction

Module 2: HTML-Introduction

- History of HTML
- What you need to do to get going and make your first HTML page
- What are HTML Tags and Attributes?
- HTML Tag vs. Element
- HTML Attributes

Module 3: HTML-Basic Formatting Tags

- HTML Basic Tags
- HTML Formatting Tags
- HTML Color Coding

Module 4: HTML-Grouping Using Div Span

- Div and Span Tags for Grouping

Module 5: HTML-Lists

- Unordered Lists
- Ordered Lists
- Definition list

Module 6: HTML-Images

- Image and Image Mapping

Module 7: HTML-Hyperlink

- URL - Uniform Resource Locator
- URL Encoding

Module 8: HTML-Table

- < table >
- < th >
- < tr >
- < td >
- < caption >
- < thead >
- < tbody >
- < tfoot >
- < colgroup >
- < col >

Module 9: HTML-Iframe

- Using Iframe as the Target

Module 10: HTML-Form

- < input >
- < textarea >
- < button >
- < select >
- < label >

Module 11: HTML-Headers

- Title
- Base
- Link
- Styles
- Script
- Meta

CSS Syllabus

Module 1: CSS2-Introduction

- Benefits of CSS
- CSS Versions History
- CSS Syntax
- External Style Sheet using < link >
- Multiple Style Sheets
- Value Lengths and Percentages

Module 2: CSS2-Syntax

- CSS Syntax
- single Style Sheets
- Multiple Style Sheets
- Value Lengths and Percentages

Module 3: CSS2-Selectors

- ID Selectors
- Class Selectors
- Grouping Selectors
- Universal Selector
- Descendant / Child Selectors
- Attribute Selectors
- CSS - Pseudo Classes

Module 4: CSS2-Color Background Cursor

- background-image
- background-repeat
- background-position
- CSS Cursor

Module 5: CSS2-Text Fonts

- color
- background-color
- text-decoration
- text-align
- vertical-align
- text-indent
- text-transform
- white space
- letter-spacing
- word-spacing
- line-height
- font-family
- font-size
- font-style
- font-variant
- font-weight

Module 6: CSS2-Lists Tables

- list-style-type
- list-style-position
- list-style-image
- list-style
- CSS Tables
  - border
  - width & height
  - text-align
  - vertical-align
  - padding
  - color

Module 7: CSS2-Box Model

- Borders & Outline
- Margin & Padding
- Height and width
- CSS Dimensions

Module 8: CSS2-Display Positioning

- CSS Visibility
- CSS Display
- CSS Scrollbars
- CSS Positioning
  - Static Positioning
  - Fixed Positioning
  - Relative Positioning
  - Absolute Positioning
- CSS Layers with Z-Index

Module 9: CSS Floats

- The float Property
- The clear Property
- The clear fix Hack

Basic JavaScript Syllabus

1. General overview of JavaScript

- What is Javascript,
- JavaScript history,
- Relation between JavaScript and ECMAScript,
- Versions of JavaScript.

JavaScript Core

1. Syntax, Variables, Values, Data Types

- Syntax review,
- Keywords and reserwed words,
- Variable declaration,
- Variable scope,
- Block scope.

2. Data Types

- Primitive values,
- Reference values,
- Types,
- Type conversion.

3. Expressions and Operators

- Expressions (arithmetic, relational, logical, assigment and others),
- Operators overview.

4. Control structures

- Flow control and conditionals,
- Loops and iteration.
- Jumps

5. Numbers

- Number literals,
- Number object,
- Number methods,
- Math object
- Date object
- Working with Numbers.

6. Strings

- String literals,
- String object,
- String methods,
- Working with Strings.

7. Arrays

- Creating and populating Arrays,
- Array methods (20+)
- Working with Arrays.

8. Functions

- Defining functions,
- Calling functions,
- Functions as values,
- Arguments and parameters,
- Function scope,
- Closures,
- Arrow functions.

9. JSON

10. Iterators and generators

- Working with Iterators
- Working with Generators

11. Modules

12. Promises

13. Regular Expressions

- Creating regular expressions,
- RegExp object and its methods,
- String methods for matching patterns.

14. Objects

- Creating object,
- Properties,
- Methods,
- Build-in JavaScript objects.

Advance JavaScript (ES6)

Section 1. New ES6 syntax
Section 2. Destructuring
Section 3. ES6 Modules
Section 4. ES6 Classes
Section 5. Arrow Functions
Section 6. Symbol
Section 7. Iterators & Generators
Section 8. Promises
Section 9. ES6 collections
Section 10. Array extensions
Section 11. Object extensions
Section 12. String extensions

Bootstrap 4 & 5 Syllabus

- Downloading the Bootstrap Files
- Understanding the File Structure
- Creating Your First Web Page with Bootstrap
- Including Bootstrap's Files via CDN
- Bootstrap Grid System
- Bootstrap Typography
- Bootstrap Tables
- Bootstrap Lists
- Bootstrap Forms
- Bootstrap Input Groups
- Bootstrap Buttons
- Bootstrap Images
- Bootstrap Media Objects
- Bootstrap Icons
- Bootstrap card
- Bootstrap Navbar
- Bootstrap Panels
- Bootstrap Breadcrumbs
- Bootstrap Pagination
- Bootstrap Labels and Badges
- Bootstrap Progress Bars
- Bootstrap Jumbotron
- Bootstrap Wells
- Bootstrap Helper Classes
- Bootstrap Modals
- Bootstrap Dropdowns
- Bootstrap Tabs
- Bootstrap Tooltips
- Bootstrap Popovers
- Bootstrap Alerts
- Bootstrap Stateful Buttons
- Bootstrap Popovers
- Bootstrap Alerts
- Bootstrap Accordion
- Bootstrap Carousel
- Bootstrap ScrollSpy
- Bootstrap Affix
- Mini Project

JQuery Syllabus

Introduction

- What You Should Already Know
- What is jQuery ?
- Adding the jQuery Library to Your Pages
- Basic jQuery Example
- Downloading jQuery
- Alternatives to Downloading
- jQuery Syntax
- The Document Ready Function
- How to use Custom Scripts?
- Using Multiple Libraries
- jQuery – noConflict() Method

jQuery – Basics

- String
- Numbers
- Boolean
- Objects
- Arrays
- Functions
- Arguments
- Scope
- Built-in Functions

jQuery – Selectors

How to use Selectors?

- jQuery – CSS Element Selector and ID Selector
- jQuery – CSS Element Class Selector and Universal Selector
- jQuery – CSS Multiple Elements E, F, G Selector
- jQuery Callback Functions

jQuery – DOM Attributes

- Get Attribute Value
- Set Attribute Value

jQuery – DOM Traversing

- Find Elements by index
- Filtering out Elements
- Locating Descendent Elements
- JQuery DOM Traversing Methods

jQuery – CSS Methods

- Apply CSS Properties and Multiple CSS Properties
- Setting Element Width & Height
- JQuery CSS Methods

jQuery – DOM Manipulation Methods

- Content Manipulation
- DOM Element Replacement
- Removing DOM Elements
- Inserting DOM elements
- DOM Manipulation Methods
- Binding event handlers
- Removing event handlers
- Event Types
- The Event Object and Attributes

jQuery – Effects

- JQuery Effect Methods, Hide and Show
- jQuery Toggle
- jQuery Slide – slideDown, slideUp, slideToggle
- jQuery Fade – fadeIn, fadeOut, fadeTo
- jQuery Custom Animations

React JS Syllabus –

Session 1 – Introduction to ReactJS

- History of front end libraries
- Motivation for using React
- Key differentiators (Virtual DOM, One way binding)
- Thinking in React

Session 2 – React Components

- React component
- Render function
- Component API
- Component lifecycle
- State
- Props
- Mixins
- JSX

Session 3 – React internals

- Reconciliation algorithm

Session 4 – Component Inter Communication

- Component composition
- Pass data from parent to child
- Pass data from child to parent

Session 5 – Component styling

- Radium
- CSS Modules
- LESS/SASS

Session 6 – Unit testing components

- Jasmine
- React test utilities

Session 7 – Performance optimizations

- PureRenderMixin
- Expensive DOM manipulations
- Performance tools

Servlet & JSP Syllabus

Servlet

Introduction

Servlet API

HTTP Protocol

Developing Servlet

Developing And Deploying Servlet Apps in Diff Servers Like

ServletRequest

ServletResponse

ServletConfig

GenericServlet

Working with welcome-file

Methods of loading Servlet

ServletContext

HttpServlet

HttpServletRequest

HttpServletResponse

RequestDispatching

Redirecting

Attributes

State and Session Management

URL Rewriting

Hidden form fields

Cookies

HttpSession

Error Handling in Servlet

Filters

Listeners

Authentication and Authorization (Security In Servlet Programming)

Working Connection Pooling

Working with domains

Annotations in Servlet Programming

JSP

8.1 Introduction

8.2 JSP Basics

8.3 JSP Tags

8.4 Scripting Elements

8.5 JSP implicit object

8.6 Directives

8.7 Standard Actions

8.8 Java Bean

8.9 Using Java Bean in JSP

8.10 Expression Language

8.11 Working JSTL

8.12 Custom Tags

8.13 MVC Architecture

8.14 Developing Project using MVC